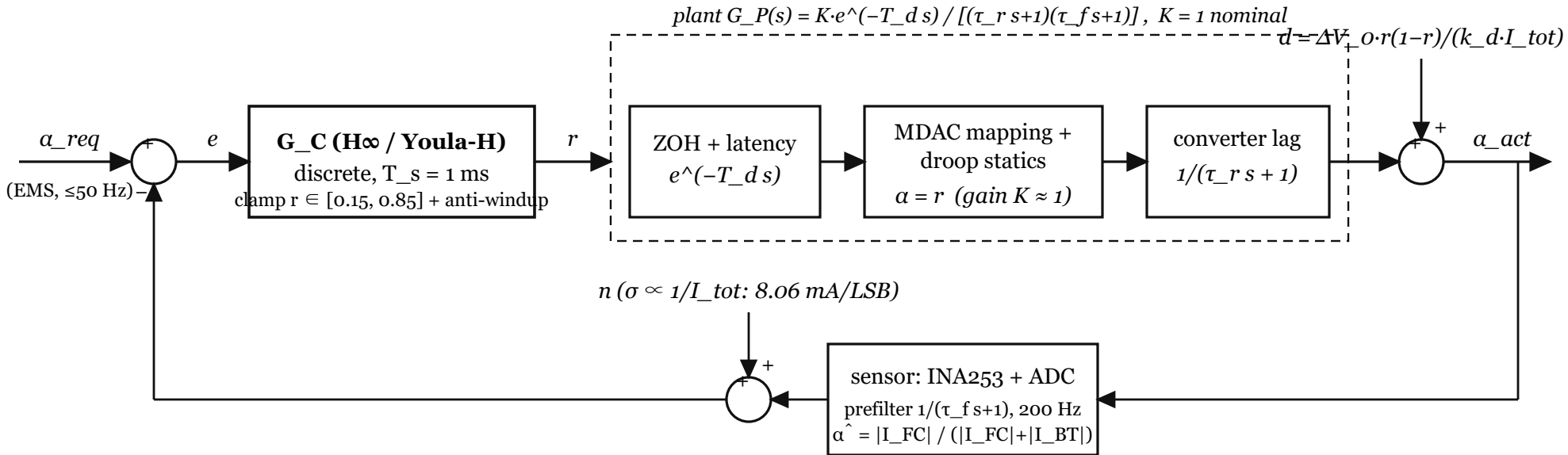


Power-share loop: linearized SISO design model



Exogenous inputs: ΔV_o (no-load setpoint mismatch, ± 0.40 V budget, $\approx$ DC) and I_{tot} (load current, VESC / drive dynamics ≤ 10 Hz) enter through d ; both are rejected at DC by enforcing $T(0) = 1$ (Youla-H). Nominal parameters: $T_d = 1.0$ ms, $\tau_r = 100$ μ s, $\tau_f = 0.8$ ms, $k_d = 0.30$ Ω .

Model validity: both sources on-bus and conducting; loop frozen when $I_{tot} \approx 0$ or BT_BUS_ENABLE low. See system_model.md §3, §6.